

venu madhav

AI/ML Engineer | Generative AI | LLM Systems

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Machine Learning Engineer with 5+ years of experience of developing software and building and deploying scalable AI/ML systems across Generative AI, multimodal learning, and LLM-based applications. Skilled in fine-tuning large language models, designing RAG and agentic workflows, and optimizing high-performance inference pipelines using PyTorch, CUDA, and TensorRT. Experienced in delivering end-to-end production AI systems, from model development to cloud deployment and real-time applications. Focused on building reliable, scalable, and production-ready AI solutions with strong business impact.

EXPERIENCE

AI/ML ENGINEER

JAN 2025 – PRESENT

PERPLEXITY

SAN FRANCISCO, CA

- Fine-tuned and deployed LLaMA 3.3–70B models using PyTorch, DeepSpeed, and HuggingFace, achieving 1,200+ tokens/sec on Cerebras CS-3 while reducing enterprise search API latency by 5×.
- Integrated AI capabilities across Labs, Finance, and Search platforms, enabling autonomous Python code generation, market reporting, and SEC-compliant insights, directly increasing Max/Pro user engagement by 27% in Q2 2025.
- Designed multi-hop RAG pipelines and grounding agents using LangGraph and FAISS, delivering accurate financial analysis and dynamic reporting with 92%+ factuality while reducing hallucinations in regulated environments.
- Developed full-stack AI workflow features using TypeScript, React (Next.js), Tailwind, and Node.js to support real-time agent interactions, prompt chaining, and AI-driven user experiences.
- Engineered scalable LLM inference services with vLLM, ONNX, and Triton on hybrid GPU/Cerebras infrastructure, improving token streaming throughput and reducing cloud infrastructure costs in production.
- Applied advanced prompt engineering and alignment techniques including RLAI, few-shot prompting, and chain-of-thought reasoning, improving performance on search, reasoning, and financial QA benchmarks by 18%.
- Built training and evaluation pipelines using Airflow, Delta Lake, HuggingFace Datasets, and Databricks, enabling rapid experimentation, model ablations, and benchmarking against GPT-4o and Claude 3.5.
- Implemented trust and safety systems using Rebuff, OpenAI Moderation API, and citation validation frameworks to ensure compliance and reliability for enterprise and financial AI workflows.
- Enhanced collaborative AI experiences through multi-user document QA and chart generation workflows, enabling business teams to co-pilot research, coding, and visualization tasks.
- Deployed and monitored production AI systems using Docker, GitHub Actions, RESTful APIs, Prometheus, and Sentry while collaborating with product and infrastructure teams to maintain scalable, low-latency services.

MACHINE LEARNING ENGINEER

NOV 2022 – JUN 2024

QUALCOMM

INDIA

- Developed Python-based multimodal perception models integrating camera, radar, and lidar data, improving obstacle detection accuracy by 38% across GPU-powered autonomous driving systems.
- Optimized deep learning inference pipelines using Python, ONNX, Qualcomm AI Engine/AI Engine Direct, Snapdragon Ride/Qualcomm AI acceleration technologies, and hardware-aware model optimization, reducing end-to-end latency by 32% for real-time automotive perception workloads.
- Built scalable active learning and preprocessing workflows in Python, increasing multimodal data pipeline efficiency by 45% and accelerating production-scale autonomous driving model training cycles.
- Designed deep learning architectures using PyTorch, TensorFlow, and ONNX for multimodal perception, occupancy estimation, and low-latency inference across embedded automotive compute environments.
- Engineered high-performance machine learning pipelines using Qualcomm AI acceleration technologies, ONNX, distributed systems, and Python, improving deployment scalability, memory utilization, and inference reliability across edge platforms.
- Developed distributed training workflows using Python, PyTorch, experiment tracking, and dataset orchestration, improving reproducibility, GPU utilization, and multimodal deep learning optimization efficiency.
- Implemented microservices-based machine learning deployment architectures using Docker, Kubernetes, and REST APIs, enabling scalable inference integration and reliable production monitoring across distributed environments.
- Deployed scalable machine learning workflows on AWS using EC2, S3, and SageMaker, enabling cloud-based training, automated evaluation, and production-ready inference deployment pipelines.

- Implemented MLOps best practices on AWS using CI/CD pipelines, model versioning, monitoring, and experiment tracking, ensuring reproducible deployments and continuous machine learning optimization.
- Collaborated with software, hardware, product, and safety engineering teams using Agile workflows and AWS infrastructure, aligning machine learning deployment with performance requirements.

SOFTWARE ENGINEER

ACCENTURE

MAY 2021 – OCT 2022

INDIA

- Assisted in developing machine learning models for credit risk scoring and underwriting using Python and Scikit-learn, improving loan default prediction accuracy by ~18% for startup-bank partnerships.
- Supported deployment of AI-powered insurance APIs by integrating computer vision models built with Python, TensorFlow, and OpenCV into cloud-hosted platforms for dynamic health risk profiling and premium calculation.
- Analyzed customer behavior data using Python, KMeans, and DBSCAN, delivering real-time personalization features that increased user engagement by 30% across fintech and insurance platforms.
- Developed RESTful AI microservices using Python, FastAPI, and Flask, deployed on Microsoft Azure with Docker and Kubernetes, and integrated with fintech systems through JSON-based APIs.
- Contributed to full-stack feature development using React.js, Python backends, PostgreSQL, and MongoDB, supporting MVP dashboards and data workflows for insurance and treasury clients.
- Assisted in building MLOps and NLP workflows using MLflow, DVC, Apache Airflow, and Hugging Face Transformers, supporting automated model versioning, experiment tracking, and retraining, while leveraging Power BI, Streamlit, GitHub Actions, and Azure DevOps for visualization and CI/CD within Agile environments.

SKILLS

Programming & Software Engineering: Python, SQL, TypeScript, JavaScript, Bash, Data Structures & Algorithms, RESTful APIs, FastAPI, Flask, Node.js, Microservices Architecture

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, Multimodal AI, NLP, Computer Vision, Predictive Modeling, Recommendation Systems, Feature Engineering, Model Optimization, Model Evaluation

Generative AI & LLMs: Large Language Models (LLMs), Generative AI, LLaMA, GPT-4o, Claude, RAG (Retrieval-Augmented Generation), Agentic AI, AI Agents, LangGraph, Prompt Engineering, Few-shot Prompting, Chain-of-Thought Reasoning, RLAIIF, Fine-tuning, Alignment, Hallucination Reduction, AI Safety & Guardrails

AI/ML Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Hugging Face Datasets, DeepSpeed, ONNX, TensorRT, OpenCV, FAISS, vLLM, CUDA, Triton Inference Server, MLflow, DVC, Apache Airflow

Cloud, MLOps & Infrastructure: AWS (EC2, S3, SageMaker), Microsoft Azure, Databricks, Delta Lake, Docker, Kubernetes, GitHub Actions, Azure DevOps, CI/CD Pipelines, Distributed Training, Model Deployment, Experiment Tracking, Monitoring, Prometheus, Sentry, Scalable AI Infrastructure

Data Engineering & Databases: Data Preprocessing, ETL Pipelines, Data Annotation, Active Learning Pipelines, Real-time Data Processing, PostgreSQL, MongoDB, Vector Databases, Data Visualization, Power BI, Streamlit

Frontend & Full-Stack Development: React.js, Next.js, Tailwind CSS, Full-Stack AI Applications, AI Workflow Development, Real-time User Interfaces, API Integration, Interactive AI Experiences, Chart & Dashboard Development

Soft Skills & Collaboration: Cross-functional Agile/Scrum Methodologies, Stakeholder Management, Research Mindset, Product Thinking, Adaptability, Team Collaboration

PROJECT

MULTIMODAL AGENTIC RAG & AI INFERENCE PLATFORM

- Architected an end-to-end multimodal RAG platform using Python, PyTorch, Hugging Face, LangGraph, and FAISS, supporting text, PDFs, images, and structured data with hybrid retrieval, reranking, tool calling, citation validation, and a target of 90%+ Recall@10 and answer faithfulness.
- Designed high-performance LLM inference using CUDA, TensorRT, ONNX, vLLM, and Triton Inference Server, applying quantization, continuous batching, and KV caching to target <1.5s P95 time-to-first-token and 100+ tokens/sec, with MLflow-based evaluation and experiment tracking.
- Productionized the platform using FastAPI, Docker, Kubernetes, AWS EC2/S3, GitHub Actions CI/CD, and Prometheus, supporting scalable REST APIs, a React/Next.js interface, and load-testing targets of 100+ requests.

EDUCATION

University of Cincinnati

Master of Science in Information Technology